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REST API **Basics**

Video 1 of "REST API Mastery" series — Easy English study notes to help you understand and explain REST APIs confidently in interviews.

Backend MERN Interview Prep

TOPIC 01

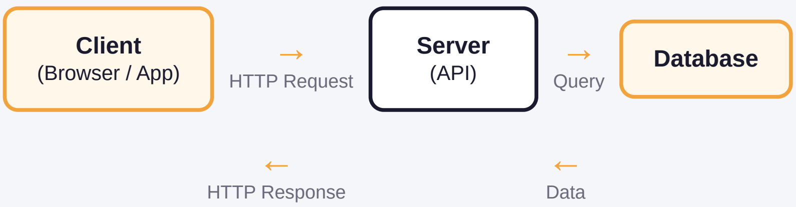
What is REST?

REST means **Representational State Transfer**. It is a set of rules (a style) to build APIs. It is not a programming language or a tool. It is just a way of designing how client and server talk to each other.

REST APIs use normal HTTP methods like GET, POST, PUT, PATCH, and DELETE to work with data. The client sends a request, and the server sends back a response.

> **Easy Analogy:** Think of REST like a restaurant. You (client) give an order to the waiter (request). The kitchen (server) prepares the food and the waiter brings it back to you (response). You don't need to know how the kitchen works inside — you just need the correct order format.

HOW REST WORKS — BASIC FLOW



> **How to answer in interview:** "REST is an architectural style for building APIs. It uses standard HTTP methods to perform operations on resources, and it follows principles like statelessness and a uniform interface."

TOPIC 02

HTTP vs REST — What's the Difference?

Many students confuse HTTP and REST. Let's make it simple.

Easy Analogy: HTTP is like the road. It is the path that lets cars (data) travel between two places. REST is like the traffic rules that decide how cars should move on that road — which lane to use, when to stop, when to go.

HTTP	REST
A protocol (a set of technical rules for sending data)	A style of designing APIs using HTTP
Defines methods like GET, POST, PUT, DELETE	Decides how to use these methods properly for resources
Works at a lower, technical level	Works at a design level, on top of HTTP

Simple line to remember: HTTP tells you that "GET /users" is a valid request. REST tells you that this request should be used to fetch the list of users.

How to answer in interview: "HTTP is the protocol that carries the request and response. REST is an architectural style that uses HTTP to structure APIs around resources."

TOPIC 03

Resource and URI

In REST, everything is treated as a **resource**. A resource is any piece of data — like a user, a product, or an order.

> **Easy Analogy:** Think of a resource like a "thing" in real life — a book in a library. Every book has its own unique shelf number so you can find it easily. In REST, that shelf number is called a **URI**.

URI means Uniform Resource Identifier. It is the address that tells the client exactly where a resource is located.

```
/api/users/123
```

This URI means: "Find the user resource whose ID is 123."

RESOURCE EXAMPLES

/users **/products** **/orders/45**

> **How to answer in interview:** "A resource is any data object like a user or a product. Each resource is uniquely identified by a URI, for example /users/123."

TOPIC 04

Path Parameters vs Query Parameters

These two are used a lot in APIs, and interviewers love asking this question.

Easy Analogy: Path parameter is like your house address — it tells "which house" exactly. Query parameter is like extra instructions you give — "please deliver after 6 PM" — it does not change the address, it just adds more detail.

```
/users/123?age=30&sort=asc
```

Path Parameter	Query Parameter
Part of the URL path	Comes after the "?" symbol
Used to identify a specific resource	Used for filtering, sorting, or pagination
Example: /users/123	Example: /users?age=30&sort=asc

Simple line to remember: Path parameter answers "WHAT to fetch." Query parameter answers "HOW to fetch it."

How to answer in interview: "Path parameters identify a specific resource, like /users/123. Query parameters are used for extra options like filtering or sorting, like /users?age=30."

TOPIC 05

Why Do We Use REST APIs?

REST is the most popular way to build APIs today. Here is why developers prefer it:

1. **Simple:** It uses normal HTTP methods, so it is easy to learn and use.
2. **Scalable:** Because REST is stateless, servers can handle many clients easily.
3. **Language-independent:** Any client (web, mobile, another server) can call a REST API, no matter what language it is written in.
4. **Flexible data format:** REST APIs usually return data in JSON, which is easy to read and use in JavaScript.
5. **Wide adoption:** Most companies use REST, so it is a must-know skill for backend developers.

Real use case: When your MERN app's frontend (React) calls the backend (Express) to fetch user data, that call is usually a REST API request.

How to answer in interview: "REST APIs are simple, stateless, and scalable. They let different clients like web and mobile apps communicate with the backend using standard HTTP methods, without depending on a specific programming language."

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Liked these notes? There's more coming.

This was Video 1 of the "REST API Mastery" series. New videos on HTTP methods, status codes, authentication, caching, and advanced REST concepts are coming soon — all explained with easy examples and interview-ready answers.

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